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1. Types of MSRs:
   1. Solid Fuel MSR (uses standard solid fuel forms, with molten salt coolant in a pressure vessel.
   2. Static Liquid Salt Reactor (liquid fuel salt, different salt used for coolant
   3. Flowing Liquid Salt Reactor (liquid fuel salt flowing in a loop, couples as a coolant

- Type 1 is only thermal spectrum, but types 2 & 3 can be both thermal or fast neutron spectrum reactors.

Benefits

- For types 2 & 3, liquid salt will dissolve fission products
- Operating pressure is ~ P_{atm}, which is 75 - 150x lower than LWR
- Higher operating temperature, leading to more efficiency electricity generation (for thermal spectrum).

2. Depending on the reactor there are different types of salts used for different purposes. Primarily there are fueled salts, and there are salts used as a coolant or thermal medium. The molten salt fuels must balance presence of fissile materials with other constituents used to either increase solubility of fission products or lower the melting temperature of the salt.
The fuel salts serve to dissolve fission products in addition to providing the primary fissile source. In flowing salt reactor loops it can sometimes also be coupled as a coolant.

3. At higher at% burnup, fluoride release affects the oxidation potential of the salt, leading to possible salt degradation and enhanced corrosion if not controlled. The cover/sparge gas helps to control the oxygen potential of the salt, and can also assist with distribution of fission products throughout the reactor.

4. The UF4/UF3 ratio is a factor used to determine the oxygen potential of the salt. As Flouride is released with higher burnup, the free electrons from the flouride are readily reactive with oxygen and can degrade the salt chemistry, affecting fission product solubility, thermal performance/characteristics of the salt, [illegible]. Chemistry of the salt must be monitored and maintained to ensure longevity and performance. The UF4/UF3 ratio is one method of determining how salt chemistry is changing.

5. Corrosion in MSRs is a major challenge due to how corrosive the molten salts are. At the beginning of life, corrosion in MSRs are driven by surface impurities. Eventually though, corrosion in-reactor is driven by a Cr-diffusion controlled reaction. One method of maintaining salt chemistry is usage of HF/H_2 to bond to free F^- ions. Structural materials in-reactor, including either steels or other nickel based alloys, have Cr as a primary constituent. The Cr and HF react to form CrF + H_2 (g). This both degrades the structural cladding and affects salt chemistry.

6. Carbide and Nitride fuels have the potential to combine the advantages of metallic and ceramic fuels. Carbide and Nitride fuels have higher melting point similar to ceramic fuels, but the thermal conductivity is much higher than UO_2 thermal conductivity.

7. Two primary pin designs are used for C and N fuels: 1) He-bonded fuel pin and 2) Na-bonded fuel pin. Generally, these systems can be summarized as follows, though some differences exist between the C and N systems.

He-bonded

- Higher smear density (~90%) to account for swelling
- Typically higher fuel centerline temp due to thermal conductivity of the He gas
- Higher fission product generation due to higher temperature

Na-bonded

- Lower smear density because swelling is higher (~85%)
- Lower fuel centerline temp due to the sodium bond + Leads to less fission product generation/release
- Na-bond inhibits gas release, causing additional swelling.

8. The three stages: Stage A, Stage B, Stage C relating to temperature evolution in C and N fuels relate strongly to fuel microstructural behavior. In Stage A, near beginning of lifetime the fuel changes are characterized by fuel cracking and fragmentation. At higher irradiation temperatures up to ~3% at burnup, the fuel resolidifies, cracks are healed, and fuel swelling occurs. Once irradiation temperature has increased sufficiently and the fuel comes in contact with the cladding, stage 3 occurs and is governed by FCCI/FCMI and creep.

9. Carbides and nitrides restructure relatively similarly into 4 primary zones. The first zone is characterized by a porous structure due to fission gas generation and release where the grain structure is changing and less stable. Zone 2 is only present in carbide fuels and only at higher temperatures at the right conditions. In this region, columnar grains aligned along the temperature gradient within the fuel form and the structure begins to become less porous. Zone 3 moving radially outward from the fuel center forms columnar grains that are coaxial, and fission bubbles are formed and moving along grain boundaries. Zone 4 is along the fuel cladding boundary, and can be represent as being nearly as-fabricated structure and porosity. As burnup increases zone one becomes more porous and moves ~~more~~ fission gases ~~to~~ the the grains boundaries in zone 3. This typically occurs during stage 3 at > 3% at burnup until end of life.

10. A major concern for FCCI in carbide fuels is cladding carbonization. At high temperatures, the carbon will react with metallic compounds in the cladding, depleting carbon in the fuel while increasing carbon content in the cladding. This significantly affects the thermal and mechanical properties of both cladding and fuel. Other intermetallic compounds can further form in the fuel, again reacting with the carbon present in the fuel. These affect fuel melting temperature and strength, potentially creating susceptible points of failure in carbide fuels.

11. The primary cause of failure in both UC and UN fuels can almost be directly traced to FCMI, swelling and the changes to mechanical and thermal properties that occur thereby. FCCI in UC and UN fuels does occur, but is far more influential in UC, rather than UN. Typically FCCI in UN has a much smaller potential to form points of susceptibility/failure. In UC fuels though, FCCI can contribute majorly due to carbonization, for temperature evolutions, stage 2 occurs at ~3% at burnup, and stage 3 occurs through end of life. In UC this could be as high as ~15% at burnup. UN fuels though can be higher than this though due to the higher density that can be achieved in fabrication

12. FCCI in UC and UN occur very similarly, aside from the carbonization of course. Due to this C/M and N/M ratios will both decrease with higher burnup. The C/M ratio especially will decrease due to carbonization. This is typically offset by increasing the carbon content during fabrication, however the C/M ratio will decrease noticeably with burnup. The N/M ratio in nitride fuels is more stable but may also decrease at higher burnup. These ratios play a key role in fuel strength and longevity, in addition to thermal properties. In short, they affect when failure might occur.